Re-measuring MMSpec: Early Results on Qwen2.5-VL-7B

Smit Thakare · Scaling LLMs final project · working note, November 9, 2026

Abstract. We re-ran the ten speculative decoding methods in the MMSpec benchmark (arXiv 2603.14989) on Qwen2.5-VL-7B under a single protocol: greedy decoding, throughput in tokens per second, and an explicit check that the output is unchanged. Acceptance lengths reproduce for eight of ten methods, which suggests the harness is measuring drafting correctly. Speed does not reproduce as cleanly. The three strongest methods are indistinguishable at roughly 1.55×, three methods emit text the target model never would, and several of the benchmark's measurement choices favour particular methods. These numbers come from 60 questions; a 600-question run is in progress.

1 Setup

Speculative decoding lets a small drafter propose several tokens which the target model then verifies in a single forward pass. Under greedy decoding the accepted prefix is exactly what the target would have produced alone, so a correct method changes speed and nothing else. We use MMSpec's released code (commit 2904868) unmodified, writing any instrumentation as separate copies. All runs use one NVIDIA L4, batch size 1, and the *testmini* split: ten questions from each of six tasks drawn from GQA, TextVQA, COCO, CharXiv, MMMU-Pro, and ConvBench/MM-MT-Bench. We report MAT, the mean number of draft tokens accepted per verification step, which depends only on the models and so transfers across hardware, and speedup as pooled tokens per second relative to a pure greedy baseline. Intervals are 95% bootstrap ranges over questions.

Method	Drafter	MAT ours / paper	Speedup ours	Speedup paper	Length vs. base	Broken answers
EAGLE-1	trained	2.35 / 2.36	1.58 (1.47–1.67)	2.11	0.99	0
ViSpec	trained, visual	2.25 / 1.29	1.57 (1.46–1.66)	1.51	1.11	0
MSD	trained, visual	2.44 / 2.57	1.56 (1.46–1.66)	2.58	0.81	8+
EAGLE-2	trained	1.78 / 1.78	1.54 (1.45–1.62)	2.02	1.01	1
Medusa	trained	0.82 / 0.80	1.12	1.49	1.06	0
PLD	copy	0.18 / 0.17	1.01 (0.95–1.07)	1.05	1.16	1
Lookahead	copy	0.34 / 0.33	0.99	1.07	1.04	0
SAM	copy	0.23 / 0.23	0.95 (0.88–1.01)	2.17	0.54	36
Recycling	copy	0.10 / 0.11	0.93	1.04	0.95	35
EAGLE-3	trained	0.05 / 0.24	0.67 (0.65–0.69)	0.96	1.06	0

Table 1. All ten methods on 60 questions. Speedup is relative to pure greedy decoding (paper figures are on A100s, ours on an L4, so only MAT is directly comparable). *Length* is total tokens generated relative to the baseline. *Broken* counts answers, out of 60, containing doubled words, junk tokens, or dropped words; the baseline has none.

2 Findings

Three methods are not lossless in the released code. SAM and Recycling produce doubled words (“are are”) or stray tokens such as *addCriterion* in more than half their answers; MSD silently drops words. Numerical noise does not explain this, since EAGLE-1 and EAGLE-3 run on the same custom model code with no such errors. Some tokens appear to be accepted without proper verification.

SAM's speedup comes from shorter output. SAM writes 0.54× as many tokens as the baseline. It finishes 1.75× sooner by wall-clock time but is no faster per token (0.95×). Its reported 6.53× on complex reasoning measures 0.99× here. This is why we report tokens per second rather than time.

The benchmark measures methods unevenly. MMSpec's baseline inherits Qwen's default repetition penalty of 1.05 while every speculative method decodes pure greedy, so the two agree on only 10% of answers; we compare against a pure greedy baseline instead. MSD starts its timer after the vision encoder has run, which excludes a median 0.86 s per turn; including it lowers its time-based speedup from 1.92× to 1.68×. MSD's per-task acceptance is also a running average over the whole run, which flattens real variation (2.16 to 2.63). Finally, the batch-size setting only reorders samples, so each request is still decoded alone and the reported batching results do not describe batched serving.

There is no clear winner. EAGLE-1, ViSpec and EAGLE-2 land between 1.54× and 1.58× with near-identical intervals. Overlap does not make them equal, but the paper's clear lead for MSD (2.58× against roughly 2.0×) does not appear. ViSpec, meanwhile, accepts 2.25 tokens per step rather than the 1.29 MMSpec reports, matching the 2.24 in the original ViSpec paper. EAGLE-3's low acceptance coincides with a warning that its drafter config loads as the wrong model type.

Even correct methods diverge late. The clean methods reproduce the baseline's exact text on only 27–42% of questions, typically splitting a fifth to a third of the way into an otherwise fluent answer. The likely cause is that verifying several tokens at once changes the order of floating-point operations, which can flip near-tied choices. Comparing each method with its own non-speculative path will separate this from genuine errors.

3 Hypotheses and next steps

Two of our four hypotheses have early answers. Drafters that see the image do not beat text-only ones on Qwen (ViSpec 1.57×, EAGLE-1 1.58×), and copy-based methods do little for vision-language models, because an answer about an image rarely repeats text from the prompt (PLD 1.01×, Lookahead 0.99×). Whether speedups survive real batched serving and how they vary with image resolution remain untested. The immediate work is to repeat every check on 600 questions with paired comparisons between methods, locate where the three faulty methods accept wrong tokens, and carry the same rules into batched serving with vLLM.

Caveats. Sixty questions is a small sample and rankings may move. Our L4 is slower than the paper's A100s, so absolute speedups are not comparable. All claims concern the released code, which may differ from what produced the published figures.